

Unit-3

Introduction to Fuzzy Logic



Fuzzy Sets

- The term fuzzy logic was first used with **1965** by Lotfi Zadeh a professor of UC Berkeley in California. He observed that **conventional computer logic was not capable of manipulating data representing subjective or unclear human ideas.**
- Fuzzy logic has been applied to various fields, from **control theory to AI.**

What is Fuzzy logic?

- Fuzzy logic is a **mathematical language** to **express** something.
 - This means it has grammar, syntax, semantic like a language for communication.
- There are some other mathematical languages also known
 - **Relational algebra** (operations on sets)
 - **Boolean algebra** (operations on Boolean variables)
 - **Predicate algebra** (operations on well formed formulae (wff), also called predicate propositions)
- **Fuzzy logic deals with **Fuzzy set** or Fuzzy algebra.**

Fuzzy logic

- **Fuzzy logic** is a form of **many-valued logic** in which the **truth values** of variables may be any real number between 0 and 1 inclusive.
- It is employed to handle the concept of partial truth, where the truth value may range between completely true and completely false.
- By contrast, in **Boolean logic**, the truth values of variables may only be the integer values 0 or 1.

- A basic application might characterize various sub-ranges of a continuous variable.
- For instance, a temperature measurement for anti-lock brakes might have several separate membership functions defining particular temperature ranges needed to control the brakes properly. Each function maps the same temperature value to a truth value in the 0 to 1 range. These truth values can then be used to determine how the brakes should be controlled.

What is Fuzzy logic?

- **The term fuzzy mean things which are not very clear or vague. In real life, we may come across a situation where we can't decide whether the statement is true or false. At that time, fuzzy logic offers very valuable flexibility for reasoning. We can also consider the uncertainties of any situation.**
- **Fuzzy logic algorithm helps to solve a problem after considering all available data. Then it takes the best possible decision for the given the input. The FL method imitates the way of decision making in a human which consider all the possibilities between digital values T and F.**

Fuzzy vs. Probability

Fuzzy : When we say about certainty of a thing

Example: A patient come to the doctor and he has to diagnose so that medicine can be prescribed.

Doctor prescribed a medicine with certainty 60% that the patient is suffering from flue. So, the disease will be cured with certainty of 60% and uncertainty 40%. Here, in stead of flue, other diseases with some other certainties may be.

Probability: When we say about the chance of an event to occur

Example: India will win the T20 tournament with a chance 60% means that out of 100 matches, India own 60 matches.

What is fuzzy?

- Dictionary meaning of **fuzzy** is **not clear, noisy**, etc.

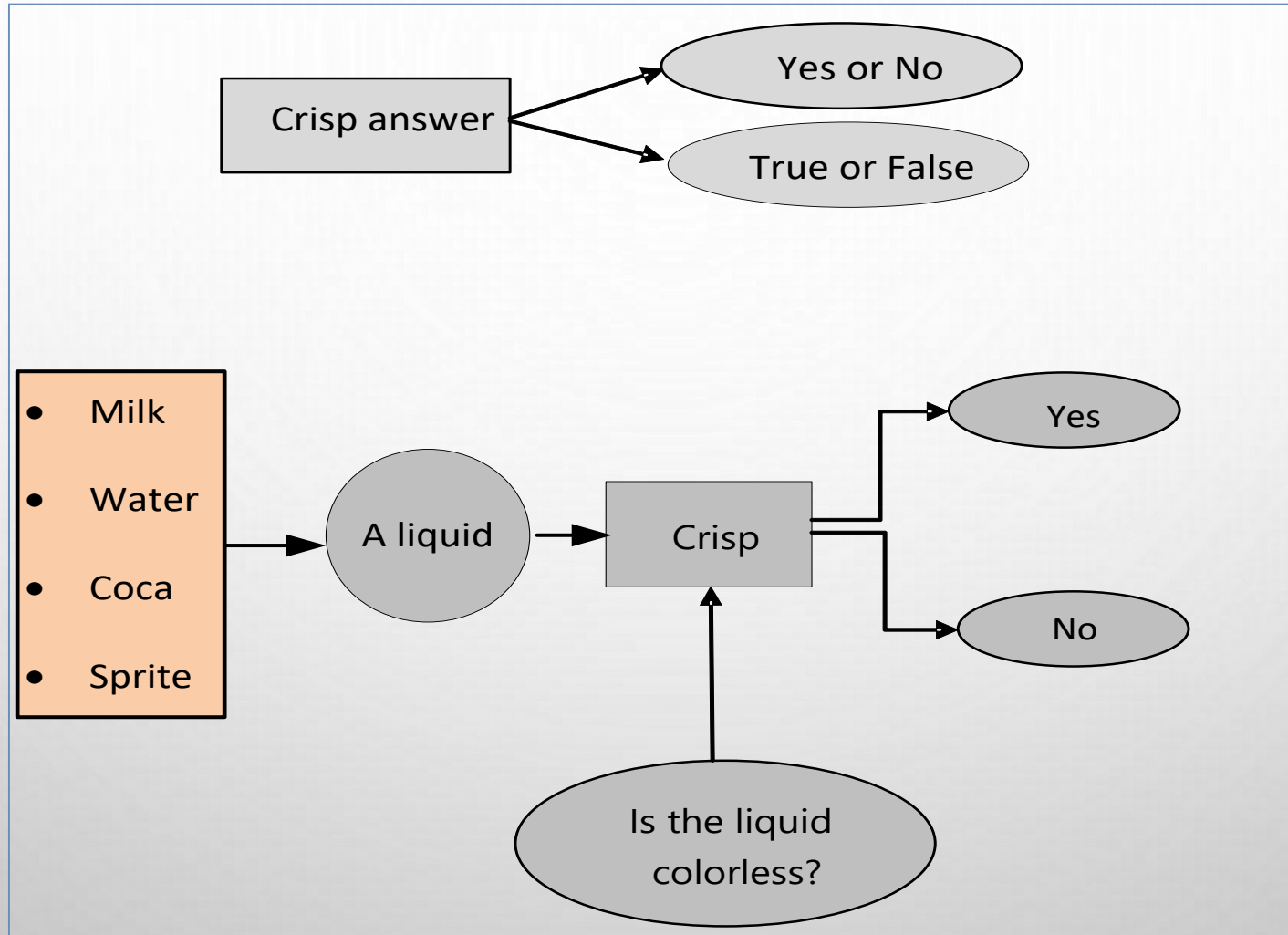
Example: Is the picture on this slide is fuzzy?

- Antonym of fuzzy is **crisp**

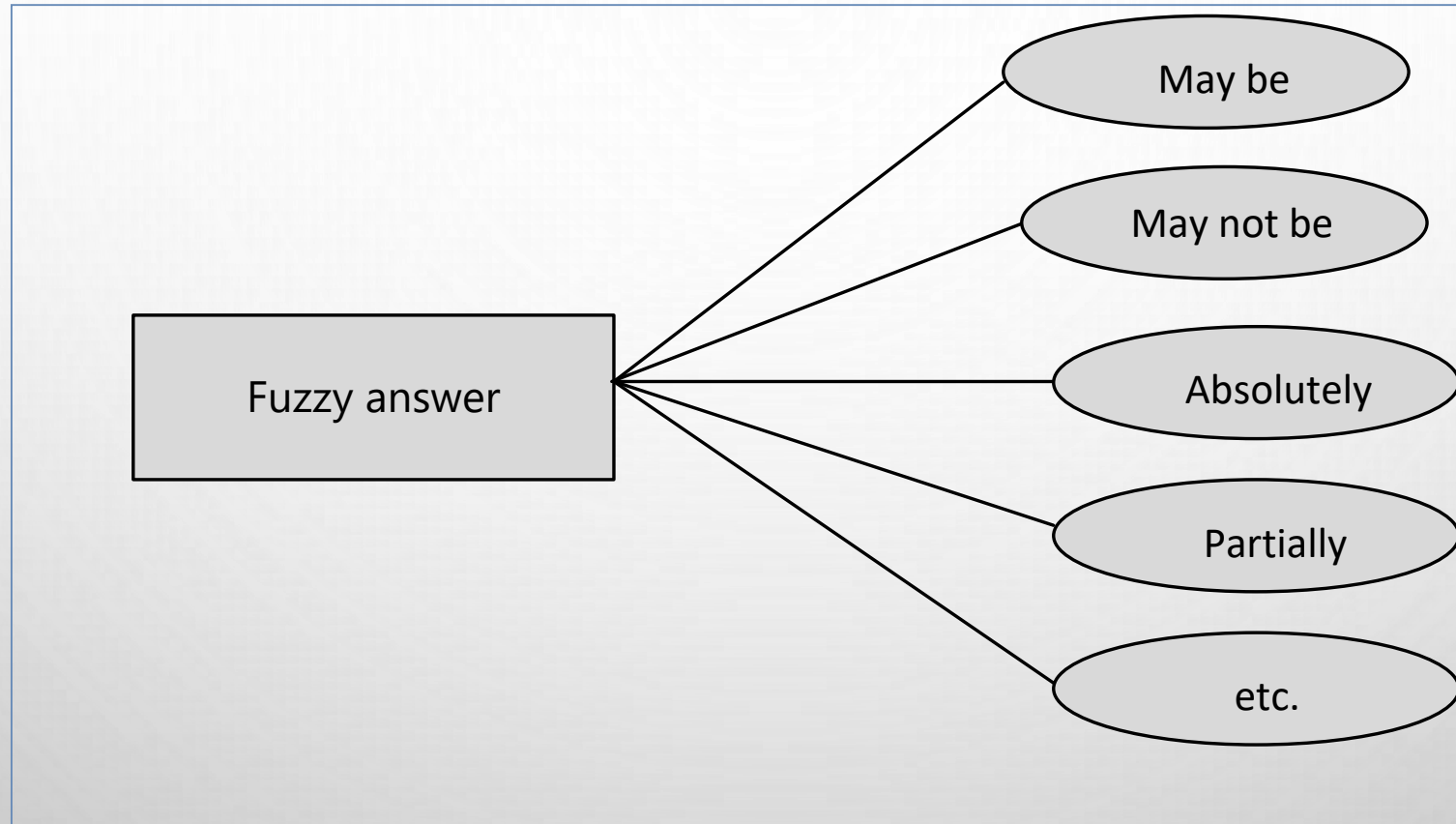
Example: Are the chips crisp?



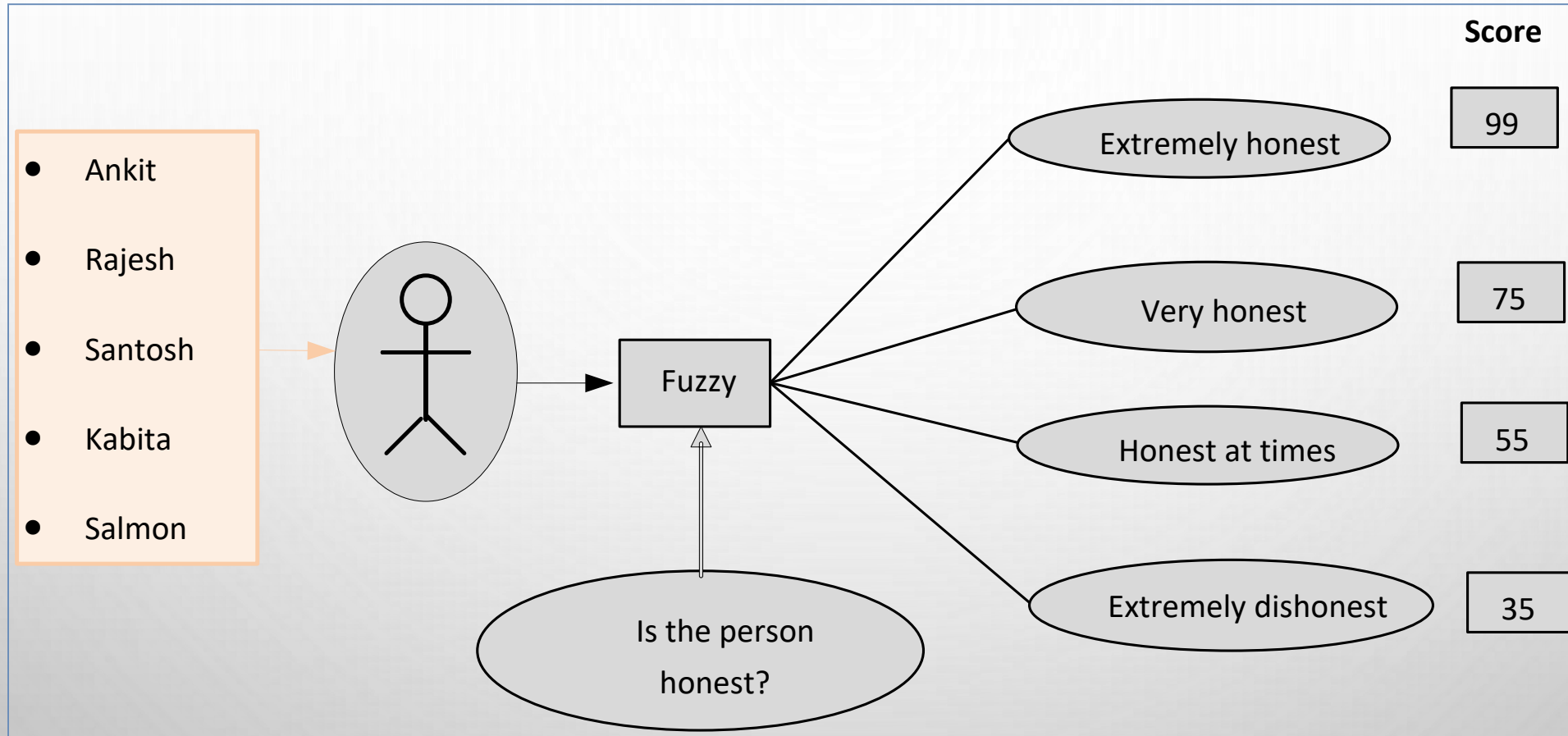
Example : Fuzzy logic vs. Crisp logic



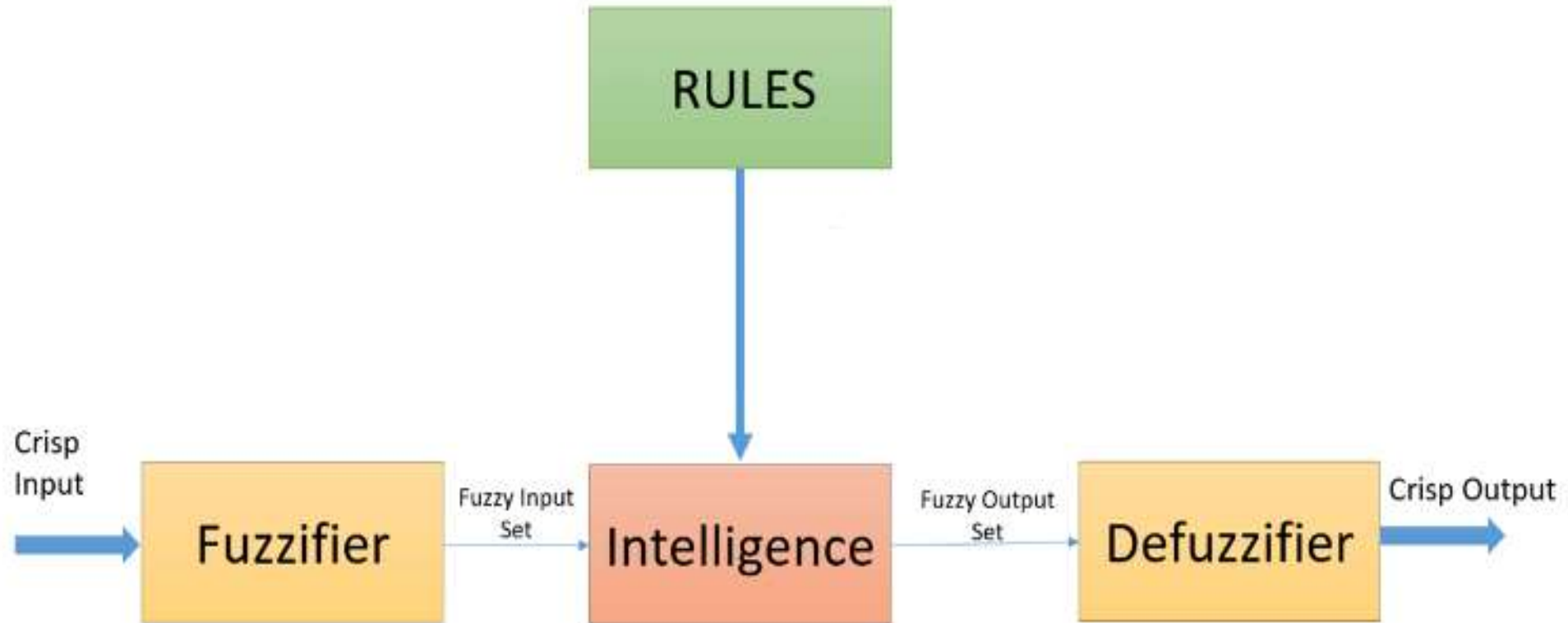
Example : Fuzzy logic vs. Crisp logic



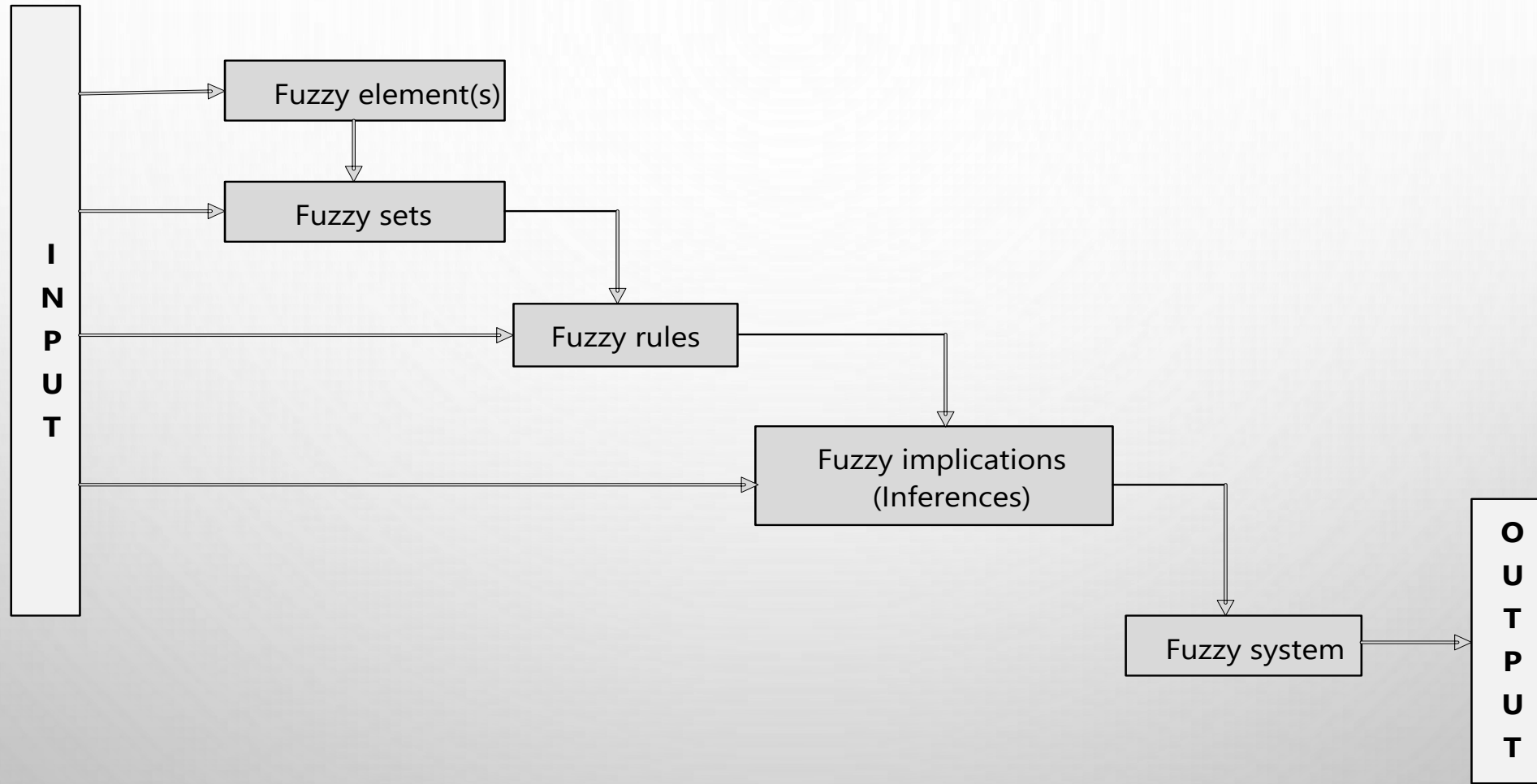
Example : Fuzzy logic vs. Crisp logic



Fuzzy Logic Architecture



Concept of fuzzy system



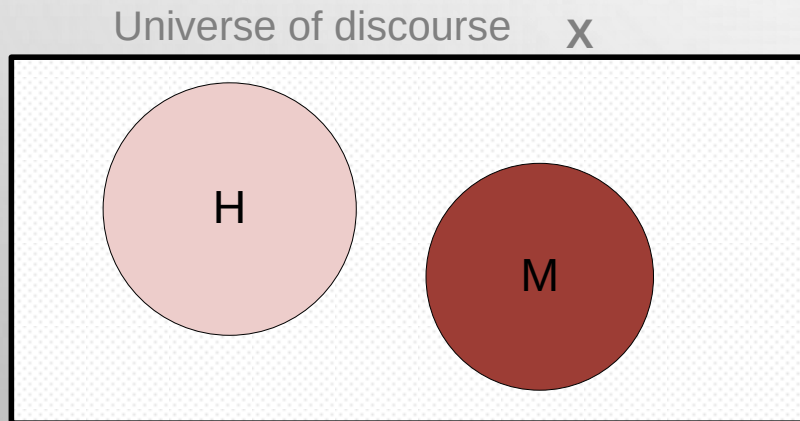
Concept of fuzzy set

To understand the concept of **fuzzy set** it is better, if we first clear our idea of **crisp set**.

X = The entire population of India.

H = All Hindu population = $\{h_1, h_2, h_3, \dots, h_L, \}$

M = All Muslim population = $\{m_1, m_2, m_3, \dots, m_N, \}$



Here, All are the sets of finite numbers of individuals. Such a set is called **crisp set**.

Example of fuzzy set

Let us discuss about fuzzy set.

X = All students in NPTEL.

S = All **Good students**.

$S = \{(s, g(s)) \mid s \in X\}$ and $g(s)$ is a measurement of goodness of the student s .

Example:

$S = \{(Rajat, 0.8), (Kabita, 0.7), (Salman, 0.1), (Ankit, 0.9)\}, \text{ etc.}$

Fuzzy set vs. Crisp set

Crisp set	Fuzzy set
▪ $S = \{s s \in X\}$	▪ $F = (s, \mu(s)) s \in X$ and $\mu(s)$ is the degree of s.
▪ It is a collection of elements.	▪ It is a collection of ordered pairs.
▪ Inclusion of an element $s \in X$ into S is crisp, that is, has strict boundary yes or no.	▪ Inclusion of an element $s \in X$ into F is fuzzy, that is, if present, then with a degree of membership.

Fuzzy set vs. Crisp set

Note: A crisp set is a fuzzy set, but, a fuzzy set is not necessarily a crisp set.

Example:

$$H = \{(h_1, 1), (h_2, 1) \dots \dots \dots, (h_L, 1)\}$$

$$\text{Person} = \{(p_1, 0), (p_2, 0) \dots \dots \dots, (p_N, 0)\}$$

In case of a crisp set, the elements are with extreme values of degree of membership namely either 1 or 0.

Degree of membership

How to decide the degree of memberships of elements in a fuzzy set?

City	Bangalore	Bombay	Hyderabad	Kharagpur	Madras	Delhi
μ	0.95	0.90	0.80	0.01	0.65	0.75

How the cities of **comfort** can be judged?

Example: Course evaluation in a crisp way

$EX : \text{Marks} \geq 90$

$A : 80 \leq \text{Marks} < 90$

$B : 70 \leq \text{Marks} < 80$

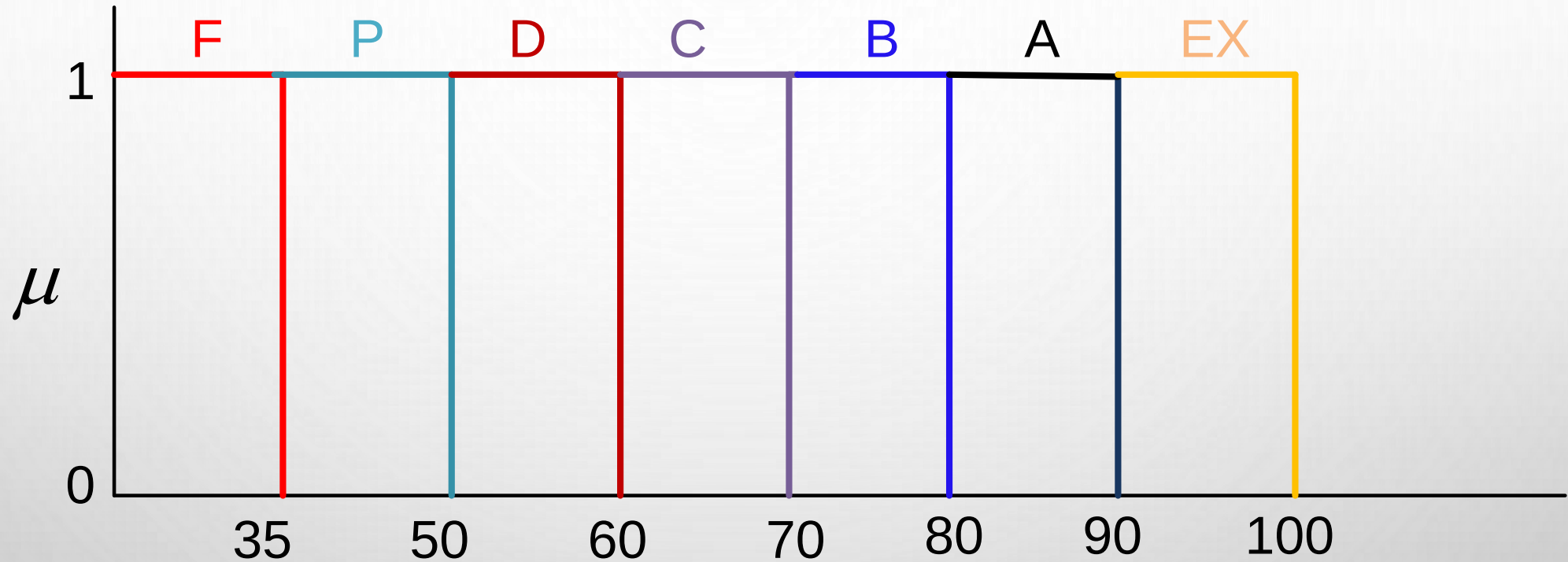
$C : 60 \leq \text{Marks} < 70$

$D : 50 \leq \text{Marks} < 60$

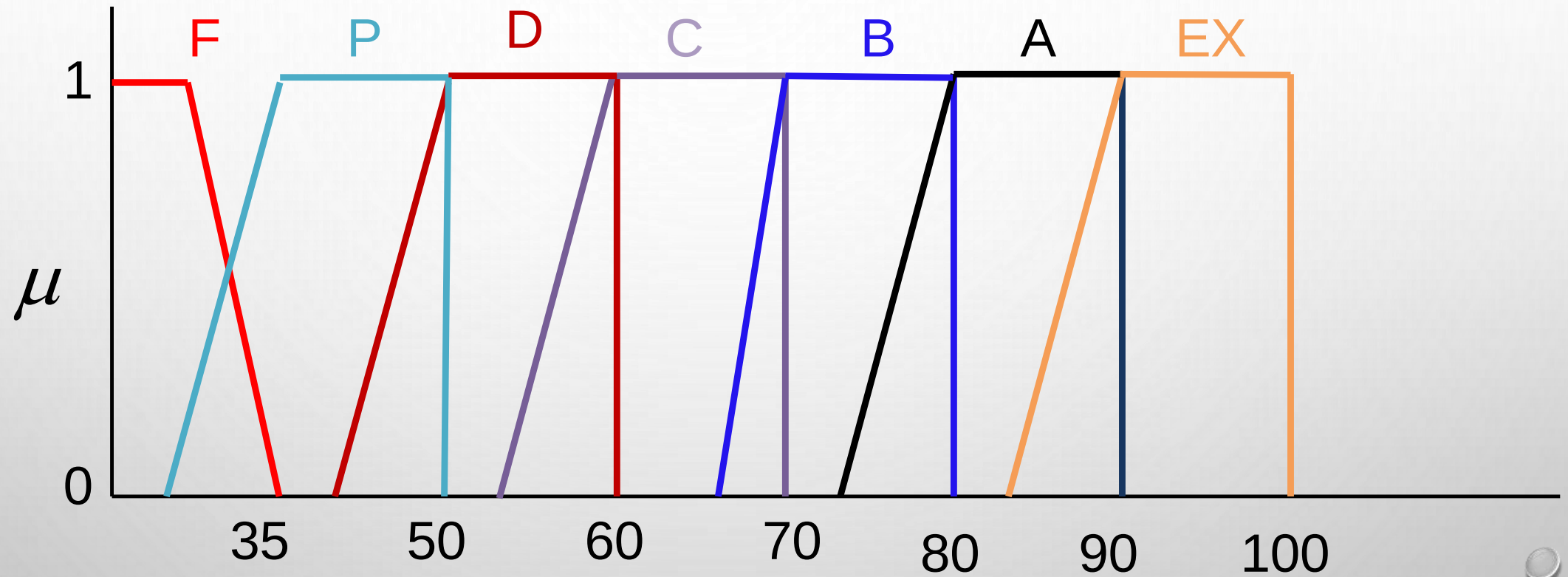
$P : 35 \leq \text{Marks} < 50$

$F : \text{Marks} \leq 35$

Example: Course evaluation in a crisp way



Example: Course evaluation in a fuzzy way



Few examples of fuzzy set

- High Temperature
- Low Pressure
- Colour of Apple
- Sweetness of Orange
- Weight of Mango

Note: Degree of membership values lie in the range $[0...1]$.

Some basic terminologies and notations

Definition 1: Membership function (and Fuzzy set)

If X is a universe of discourse and $x \in X$, then a fuzzy set A in X is defined as a set of ordered pairs, that is

$A = \{(x, \mu_A(x)) | x \in X\}$ where $\mu_A(x)$ is called the **membership function** for the fuzzy set A .

Note: $\mu_A(x)$ map each element of X onto a membership grade (or membership value) between 0 and 1 (both inclusive).

Question: How (and who) decides $\mu_A(x)$ for a fuzzy set A in X ?

Some basic terminologies and notations

Example:

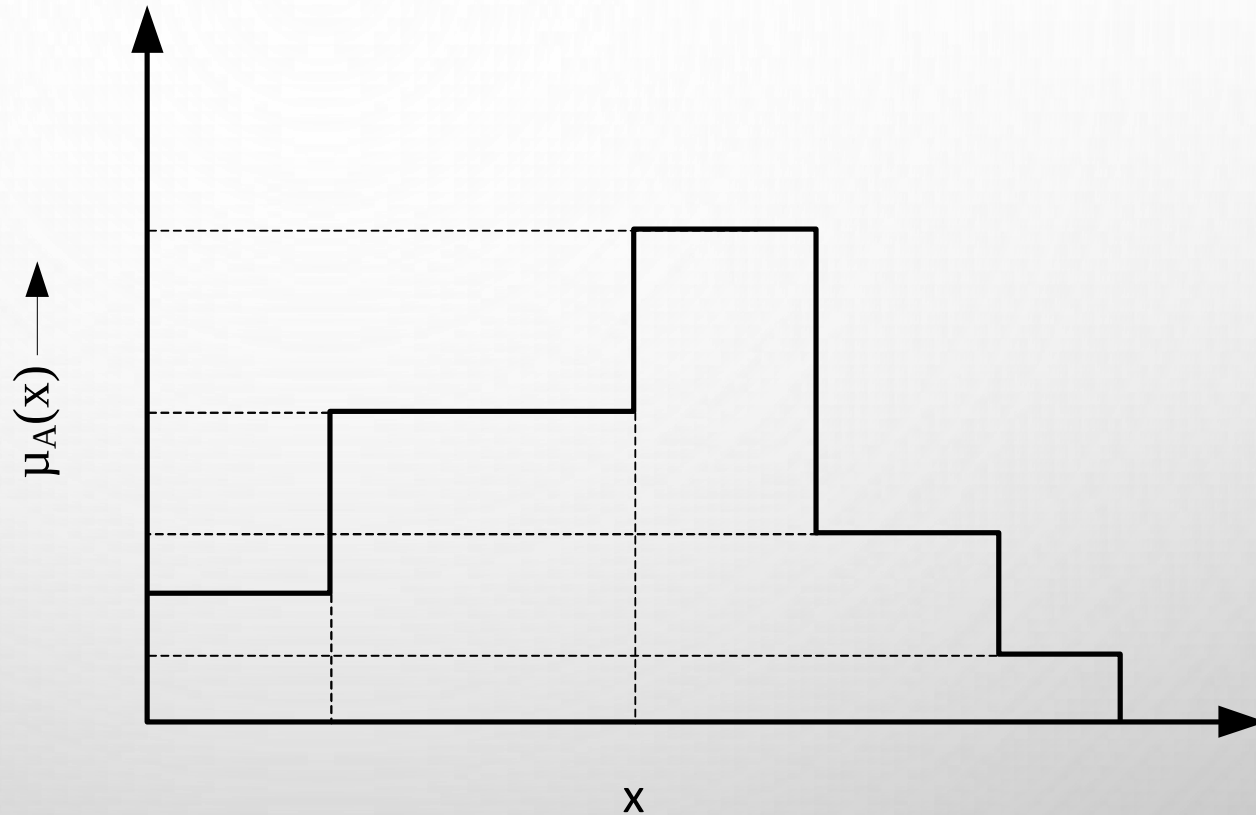
X = All cities in India

A = City of comfort

$A = \{(\text{New Delhi}, 0.7), (\text{Bangalore}, 0.9), (\text{Chennai}, 0.8), (\text{Hyderabad}, 0.6),$
 $(\text{Kolkata}, 0.3), (\text{Kharagpur}, 0)\}$

Membership function with discrete membership values

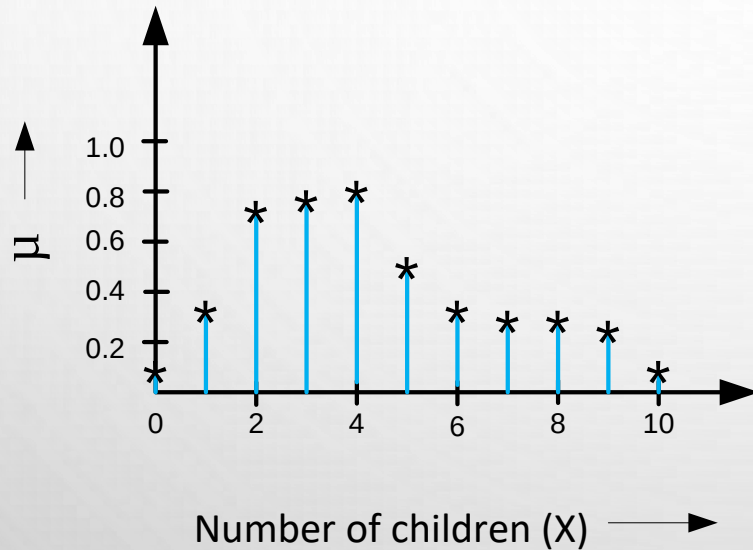
The membership values may be of discrete values.



A fuzzy set with discrete values of μ

Membership function with discrete membership values

Either elements or their membership values (or both) also may be of discrete values.



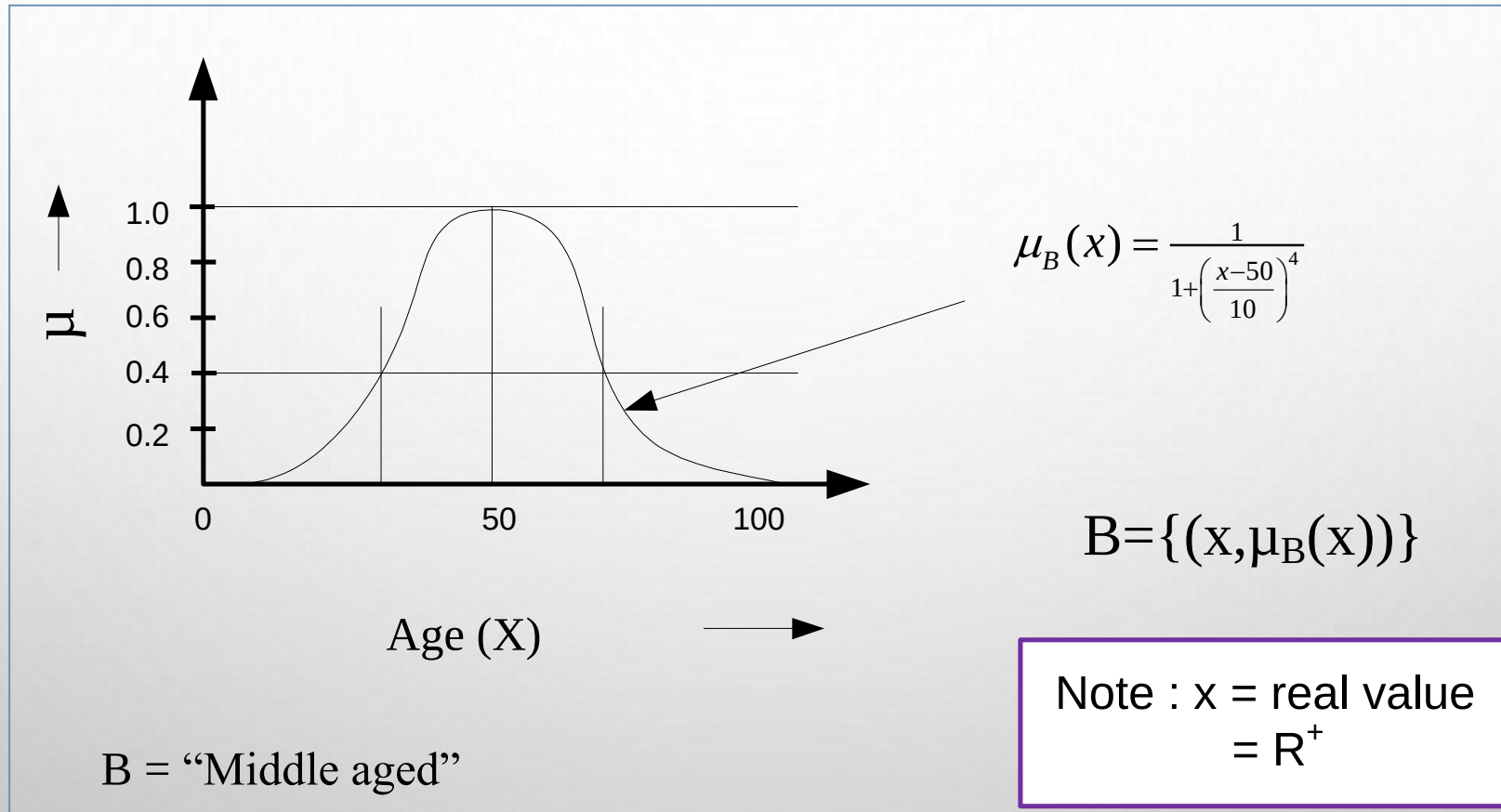
A = "Happy family"

$$A = \{(0, 0.1), (1, 0.30), (2, 0.78), \dots, (10, 0.1)\}$$

Note : X = discrete value

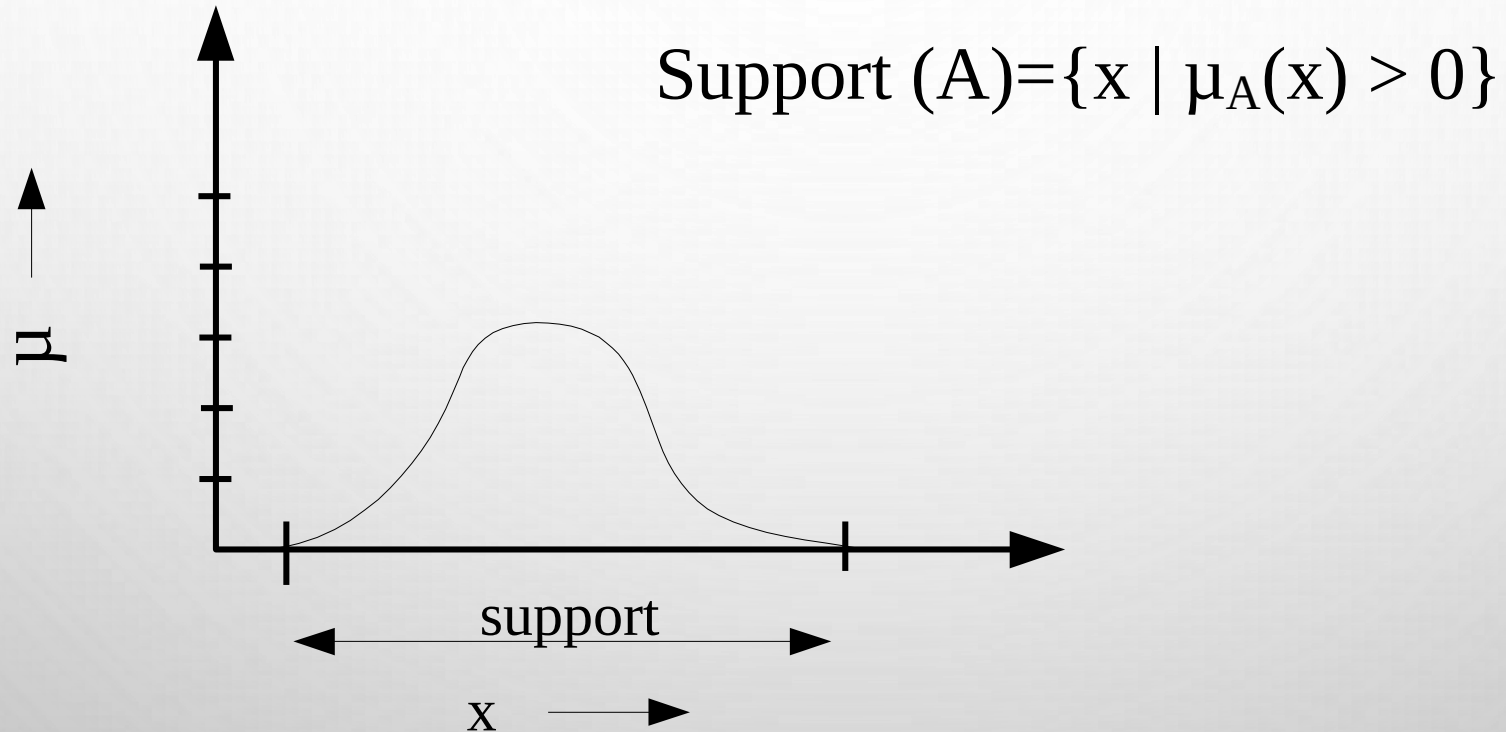
How you measure happiness ??

Membership function with continuous membership values



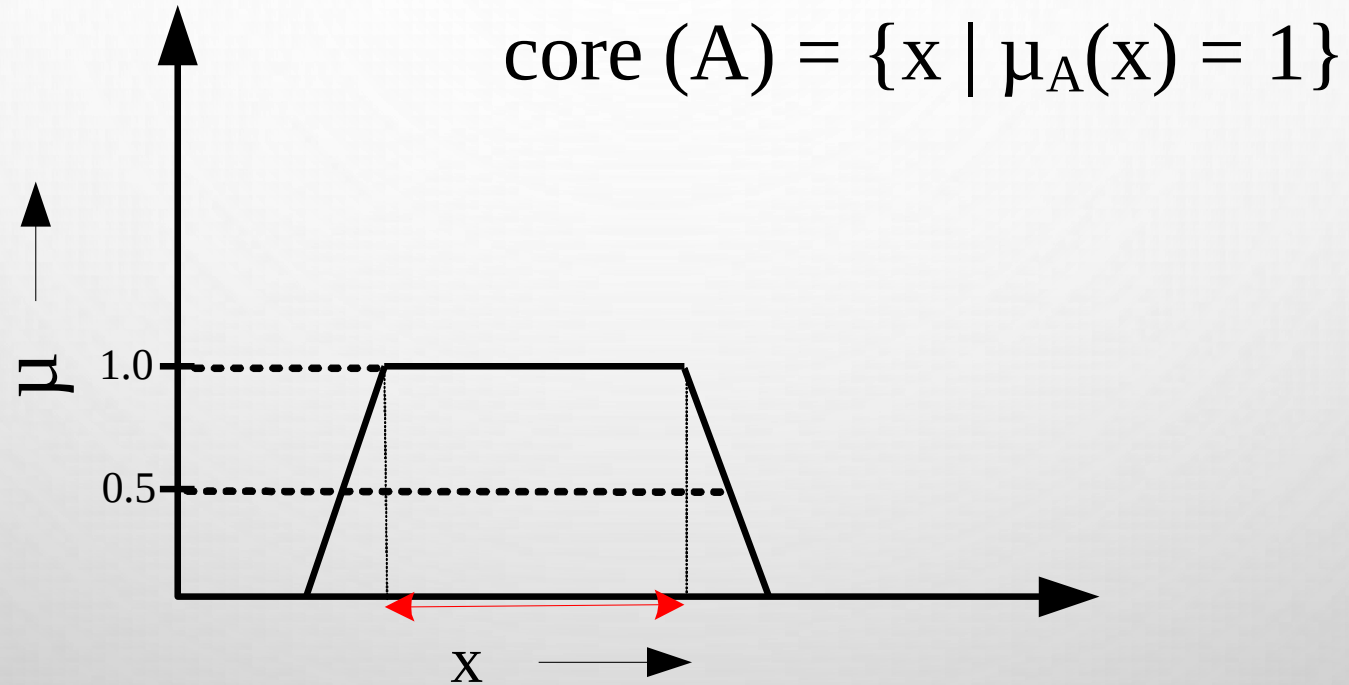
Fuzzy terminologies: Support

Support: The support of a fuzzy set A is the set of all points $x \in X$ such that $\mu_A(x) > 0$



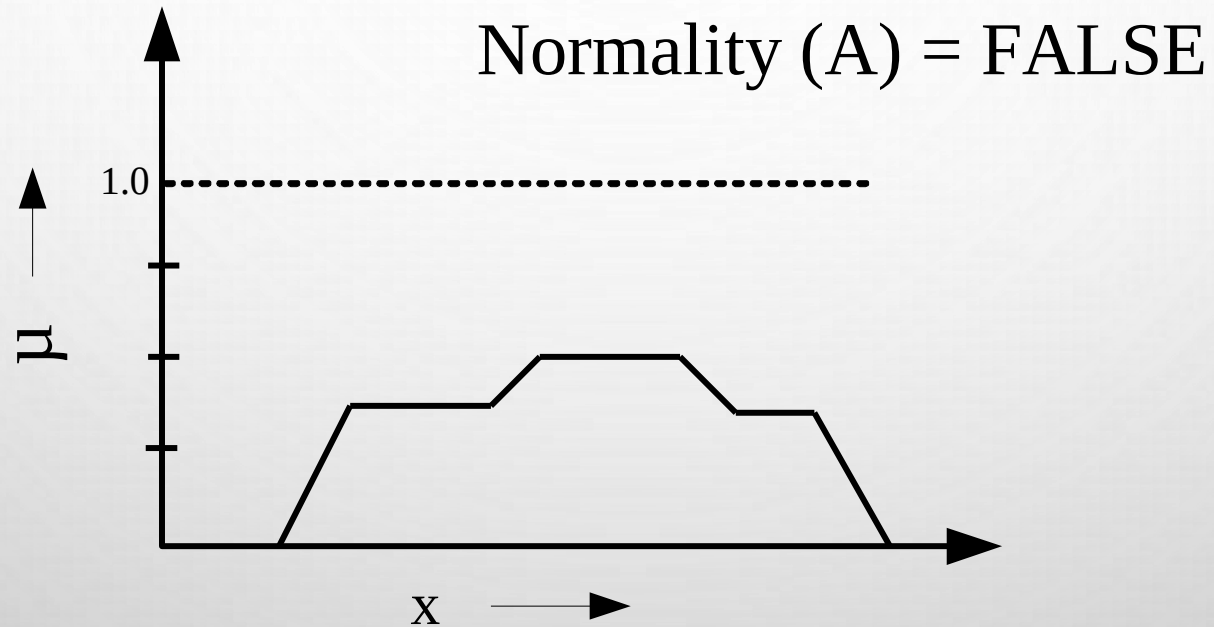
Fuzzy terminologies: Core

Core: The core of a fuzzy set A is the set of all points x in X such that $\mu_A(x) = 1$



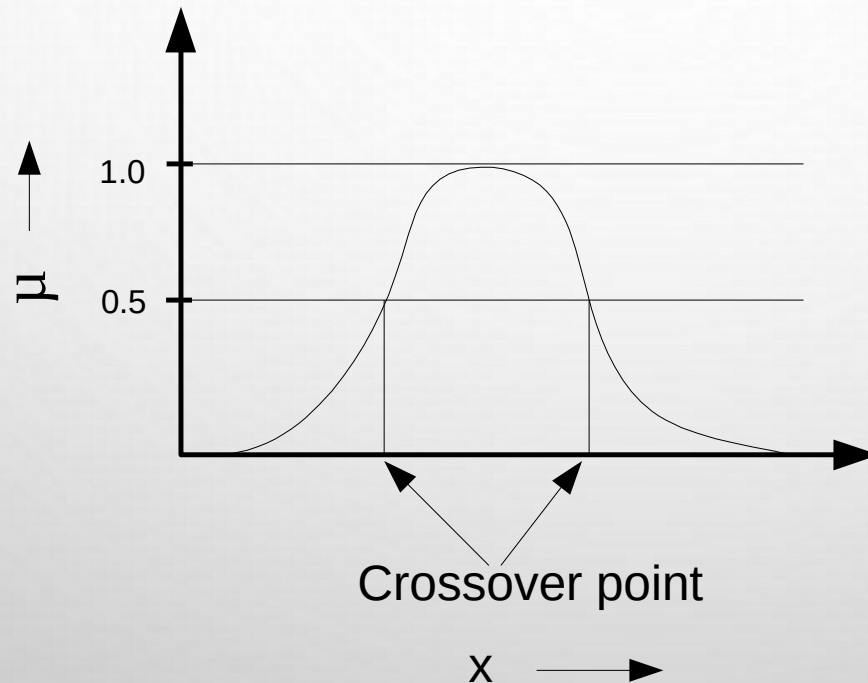
Fuzzy terminologies: Normality

Normality : A fuzzy set A is a normal if its core is non-empty. In other words, we can always find a point $x \in X$ such that $\mu_A(x) = 1$



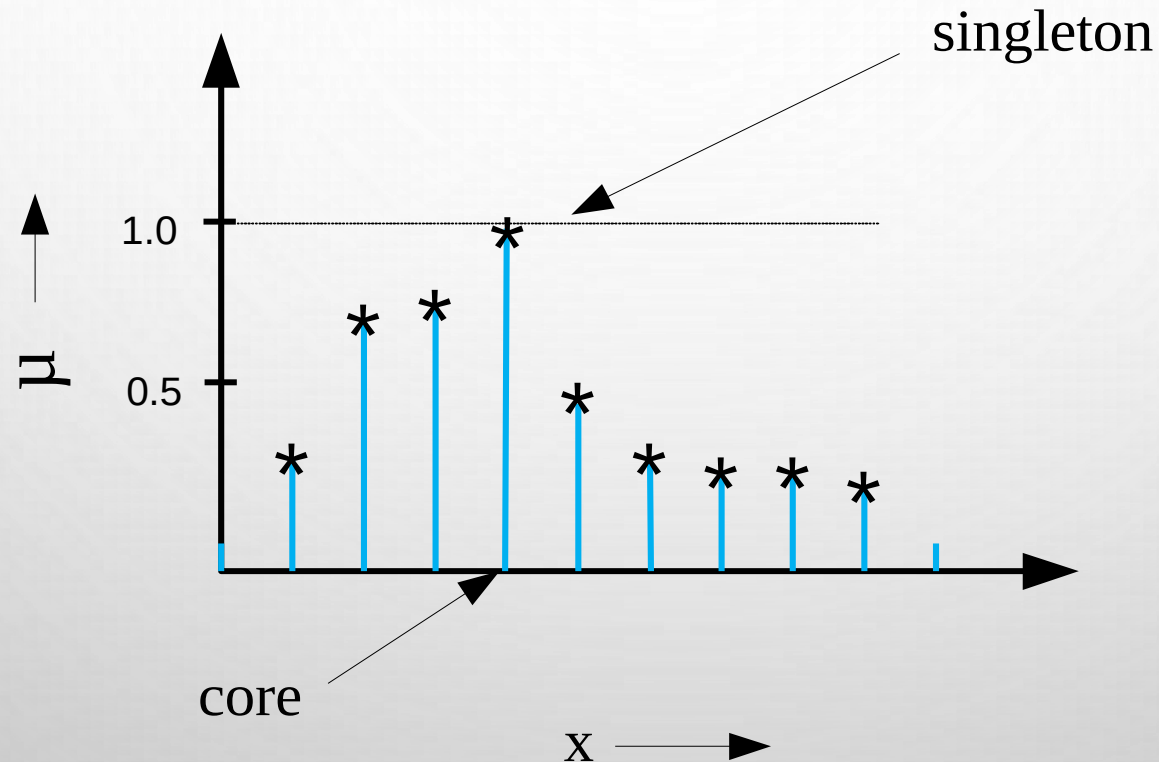
Fuzzy terminologies: Crossover points

Crossover point : A crossover point of a fuzzy set A is a point $x \in X$ at which $\mu_A(x) = 0.5$. That is $\text{Crossover}(A) = \{x | \mu_A(x) = 0.5\}$



Fuzzy terminologies: Fuzzy Singleton

Fuzzy Singleton : A fuzzy set whose support is a single point in X with $\mu_A(x) = 1$ is called a fuzzy singleton. That is $|A| = \{x | \mu_A(x) = 1\}$



Fuzzy terminologies: α -cut and strong α -cut

α -cut and strong α -cut :

- ✓ The α -cut of a fuzzy set A is a crisp set defined by

$$A_{\alpha} = \{x | \mu_A(x) \geq \alpha\}$$

- ✓ Strong α -cut is defined similarly :

$$A'_{\alpha} = \{x | \mu_A(x) > \alpha\}$$

Note : Support (A) = A_0 ' and Core (A) = A_1 .

Fuzzy terminologies: Bandwidth

Bandwidth :

For a fuzzy set, the bandwidth (or width) is defined as the distance between the two unique crossover points:

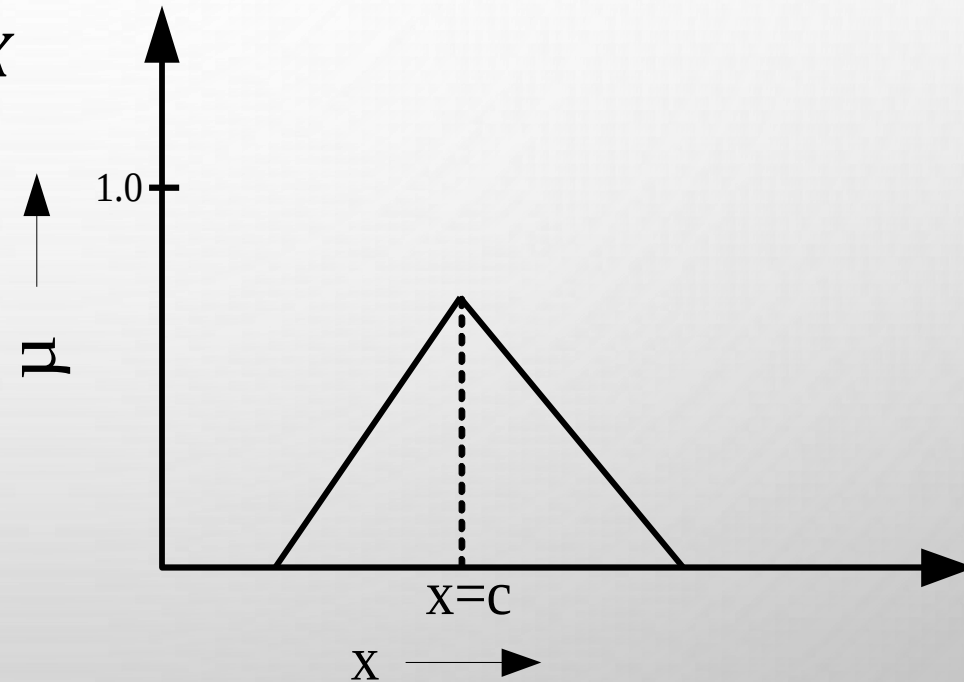
$$\text{Bandwidth } (A) = |x_1 - x_2|$$

where $\mu_A(x_1) = \mu_A(x_2) = 0.5$

Fuzzy terminologies: Symmetry

Symmetry :

A fuzzy set A is symmetric if its membership function around a certain point $x = c$, namely $\mu_A(x + c) = \mu_A(x - c)$ for all $x \in X$



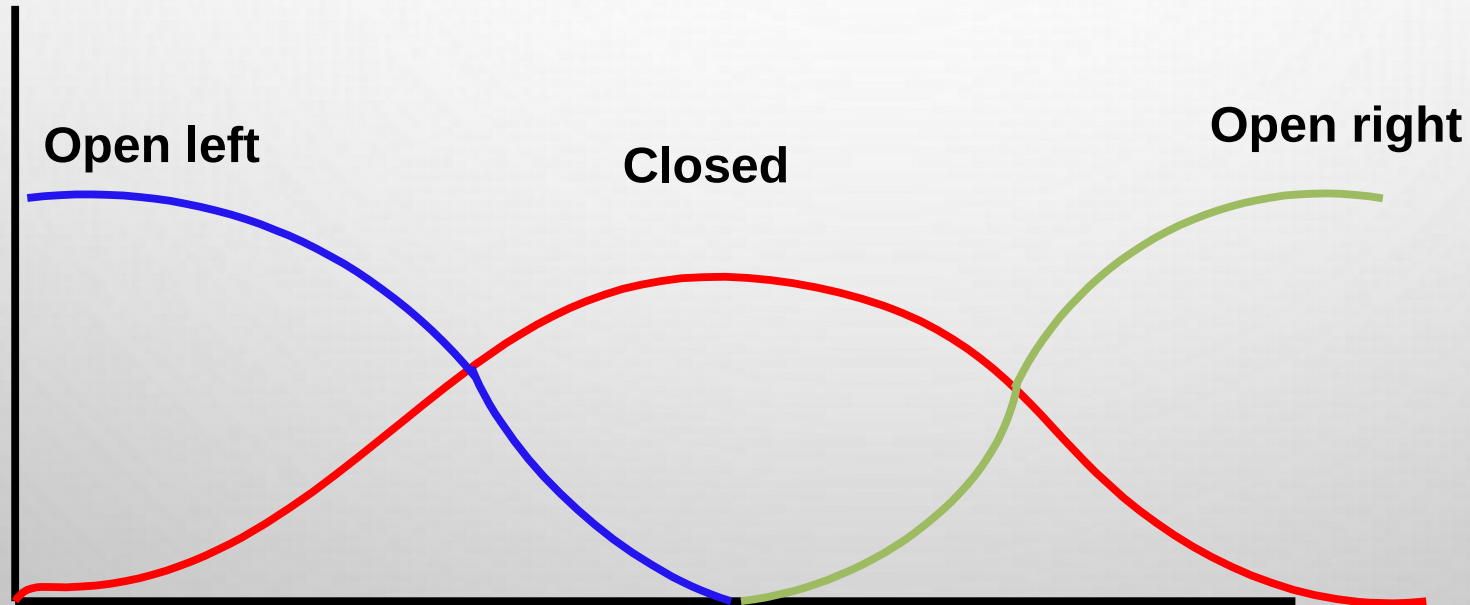
Fuzzy terminologies: Open and Closed

A fuzzy set A is

Open left : If $\lim_{x \rightarrow -\infty} \mu_A(x) = 1$ and $\lim_{x \rightarrow +\infty} \mu_A(x) = 0$

Open right: If $\lim_{x \rightarrow -\infty} \mu_A(x) = 0$ and $\lim_{x \rightarrow +\infty} \mu_A(x) = 1$

Closed: If $\lim_{x \rightarrow -\infty} \mu_A(x) = \lim_{x \rightarrow +\infty} \mu_A(x) = 0$



Fuzzy vs. Probability

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Prediction vs. Forecasting

The Fuzzy vs. Probability is analogical to Prediction vs. Forecasting

Prediction : When you start guessing about things.

Forecasting : When you take the information from the past job and apply it to new job.

The main difference:

Prediction is based on the best guess from experiences.

Forecasting is based on data you have actually recorded and packed from previous job.